

Assignment: Control-Constrained Optimal Control
Semismooth Newton / Primal-Dual Active Set Strategy

1 Problem description

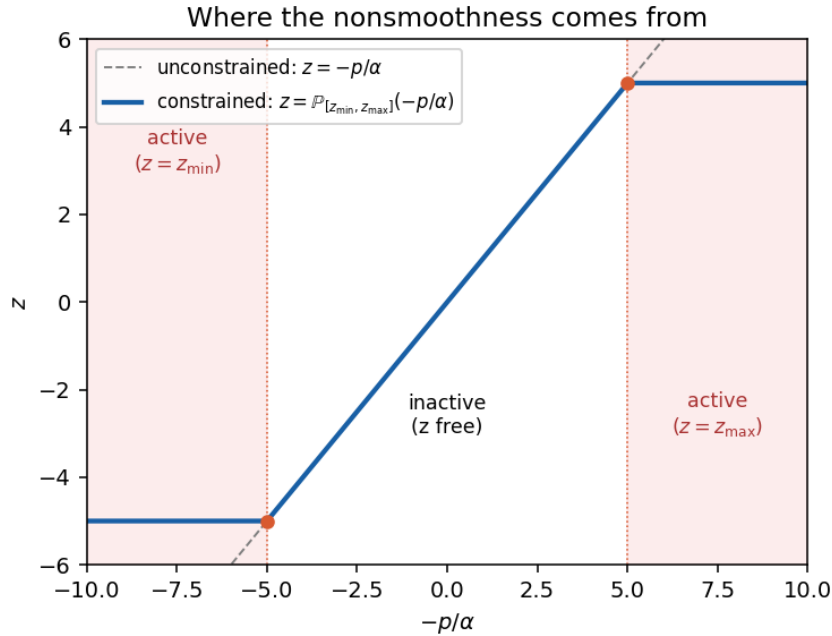
Consider a distributed-control problem: choose a source z so that the resulting state u tracks a target u_d , penalizing $\|z\|$,

$$\min_{u,z} \frac{1}{2} \int_{\Omega} (u - u_d)^2 + \frac{\alpha}{2} \int_{\Omega} z^2 \quad \text{s.t.} \quad -\Delta u = z, \quad u = 0 \text{ on } \partial\Omega,$$

but now with pointwise bounds on the control:

$$z_{\min} \leq z(x, y) \leq z_{\max} \quad \text{a.e. in } \Omega$$

Without bounds, the optimal control is given by a smooth formula, $z = -p/\alpha$ (with p the adjoint state). With bounds, whenever this formula would leave $[z_{\min}, z_{\max}]$, z is clipped to the nearest bound instead. This introduces a kink: the map from p to z is continuous but not differentiable at the two points where it starts clipping.



This is the sense in which the problem becomes *nonsmooth*. A plain gradient-based iteration still works but converges slowly near the kink; the goal of this assignment is to implement a scheme that instead treats the two sides of the kink exactly.

2 Governing equations

Optimality system. A minimizer of $J(u, z)$ subject to $-\Delta u = z$, $u = 0$ on $\partial\Omega$, and $z_{\min} \leq z \leq z_{\max}$, satisfies:

$$-\Delta u = z, \quad -\Delta p = u - u_d, \quad z = \mathbb{P}_{[z_{\min}, z_{\max}]} \left(-\frac{p}{\alpha} \right)$$

(all with homogeneous Dirichlet boundary conditions on u, p), where $\mathbb{P}_{[z_{\min}, z_{\max}]}$ denotes pointwise clipping to the interval.

Primal-dual active set system. Given a partition of Ω into an active set $\mathcal{A} = \mathcal{A}^+ \cup \mathcal{A}^-$ (where the bound is attained) and an inactive set $\mathcal{I}$ (where z is free), with S the diagonal indicator of $\mathcal{I}$ and z_{bd} holding the bound value on $\mathcal{A}$ (zero on $\mathcal{I}$), the discrete system to solve at each active-set iteration is:

$$\begin{pmatrix} A & \frac{1}{\alpha}MS \\ M & -A \end{pmatrix} \begin{pmatrix} u \\ p \end{pmatrix} = \begin{pmatrix} M z_{\text{bd}} \\ M u_d \end{pmatrix}$$

The active set is then updated from the new p ($\mathcal{A}^+ = \{i : -p_i/\alpha > z_{\max}\}$, $\mathcal{A}^- = \{i : -p_i/\alpha < z_{\min}\}$), and the process repeats until the partition stops changing.

3 Task

1. Implement a **baseline**: projected gradient descent, $z \leftarrow \mathbb{P}_{[z_{\min}, z_{\max}]}(z - \tau g)$ with $g = \alpha z + p$.
2. Implement the **primal-dual active set method** using the coupled system above, iterating until the active set is unchanged between two consecutive iterations.
3. Verify that the active set partition stabilizes completely after finitely many iterations (unlike the baseline, which never exactly stops changing).
4. **Mesh-independence test.** Solve with both methods at several mesh resolutions. Plot the number of outer iterations to convergence against mesh resolution, for both methods, on the same axes. The primal-dual method's iteration count is expected to stay roughly flat; the projected gradient's is expected to grow.

4 Data

Symbol	Value	Meaning
Ω	$(0, 1)^2$	domain
$u_d(x, y)$	$\sin(\pi x) \sin(\pi y)$	target state
α	10^{-2}	regularization weight

Choosing $z_{\min}, z_{\max}$. First solve the same problem *without* the box constraint (i.e. using only $-\Delta u = z$, $-\Delta p = u - u_d$, $\alpha z + p = 0$) at this α , and inspect the range of the resulting z . Then pick $z_{\min}, z_{\max}$ strictly inside that range, so that the constraint is actually active on part of Ω — otherwise the exercise validates nothing.

Mesh resolutions for the independence test: at least three increasing resolutions (e.g. doubling the number of elements per side each time).